

PRACTICE Qs VIDEO: Q2 - LC 268 MISSING NUMBER

GIVEN AN ARRAY "NUMS" CONTAINING N DISTINCT NUMBERS IN THE RANGE $[0, N]$
RETURN THE ONLY NUMBER IN THE RANGE MISSING FROM THE ARRAY

EX: NUMS = [3, 0, 1] RETURN 2

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THIS IS SORT OF TRICKY. WE INPUT NUMS = [0, 1] SO WE HAVE $n=2$ ELEMENTS
BUT OUR RANGE IS $[0, N]$ OR $[0, 2]$

↳ SO OUR RANGE IS ALWAYS 1 BIGGER THAN OUR ARRAY SO THERE WILL ALWAYS BE 1 MISSING

THIS WILL MAKE IT $O(n \log n) + O(n)$

IF WE SORT THE ARRAY IT SHOULD INCREMENT BY 1 UNLESS THAT VALUE IS MISSING

EX: CASE 1: LEFT EDGE NUMS = [3, 2, 1] $\rightarrow$ RANGE $[0, 3]$

SORT NUMS = [1, 2, 3]

*WE CAN ALWAYS START AT 0 SINCE RANGE STARTS AT 0

WE CAN USE A FIXED SLIDING WINDOW OF SIZE 2 KEEPS THIS $O(n)$

0 [1, 2, 3] BUT I THINK FOR THIS EDGE CASE WE SHOULD JUST CHECK IF
THE FIRST ELEMENT IS 0, IF NOT THEN THAT IS THE
ONLY POSSIBLE MISSING VALUE.

CASE 2: MIDDLE NUMS = [0, 2, 3] $\rightarrow$ RANGE $[0, 3]$

$i \quad i+1$
 $[i+1] - [i] \geq 2 > 1$ SO THE VALUE BETWEEN $[i+1]$ & $[i]$ IS
MISSING. THIS VALUE IS $[i] + 1$

CASE 3: RIGHT EDGE NUMS = [0, 1, 2] $\rightarrow$ RANGE $[0, 3]$

SINCE WE KNOW THERE IS ALWAYS 1 MISSING

IF CASE 1 OR CASE 2 PASS THEN THE ONE MISSING MUST BE THE LAST
VALUE WHICH IS n . SO IF WE EXIT OUR LOOP WITHOUT TRIGGERING
CASE 1 OR 2 THEN WE JUST RETURN n .

WHAT IF WE DIDN'T SORT? IS IT POSSIBLE? (THIS IS $O(n) + O(n)$ TIME)
WE COULD ITERATE THROUGH EACH NUM AND MAP IT AS 1 OR 0 IN AN ARRAY AT ITS INDEX
THEN GO THROUGH THAT ARRAY & RETURN THE INDEX WITH A 0 (THIS ACCOUNTS FOR THE LEFT EDGE)
IF THE ARRAY IS ALL 1'S THEN THE MISSING VALUE IS THE RIGHT EDGE. RETURN n